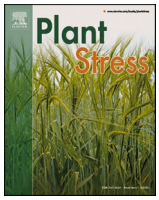
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Arbuscular mycorrhizal fungi induce Zn uptake and antioxidant efficiency in broccoli (*Brassica oleracea* L.) exposed to Zn deficiency☆

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ABSTRACT

Zinc (Zn) is an essential micronutrient for plant growth and development, and its deficiency in soil can be a significant problem for broccoli (*Brassica oleracea* L.) production. While the genetic efforts to breed Zn-efficient broccoli are complex, harnessing microbiomes is an emerging way to induce mineral efficiency in crops. In this study, broccoli, a non-host mycorrhizal species, was cultivated with or without zinc deficiency in the presence or absence of arbuscular mycorrhizal fungi (AMF), confirmed by the colonization efficiency in roots. Zn deficiency caused a significant decrease in morphological parameters and photosynthetic attributes being consistent with the decreased Zn levels in root and shoot relative to Zn-sufficient plants. However, the broccoli plants inoculated with AMF showed a substantial improvement in morphological and photosynthetic parameters in Zn-deficient conditions due to the significant increase in tissue Zn levels. In addition, AMF-inoculated plants under Zn deficiency showed a significant decrease in cell death (%), electrolyte leakage, and H_2O_2 levels compared to Zn-deficient plants, which further suggests the improvements in cellular status due to the AMF colonization. The real-time PCR experiments showed a significant induction in the expression of *BoZIP1* and *BoNRAMP1* genes in the roots of Zn-deficient broccoli inoculated with AMF, suggesting that AMF may be associated with the induction of these genes responsible for Zn-uptake and transport. We further observed the induction of POD (peroxidase), APX (ascorbate peroxidase) and SOD (superoxide dismutase), and S-metabolites (cysteine and glutathione) predominantly in the roots of Zn-starved broccoli inoculated with AMF which may confer tolerance to Zn-deficiency induced oxidative damage. This is the first report on the role of AMF in mitigating Zn-deficiency in broccoli which may promote the microbiome-aided improvement in plant health suffering from Zn-deficiency in broccoli and other vegetable crops.

1. Introduction

Zn (Zn) is an essential micronutrient for plant growth and development, and its deficiency in soil can be a significant problem for farmers and breeders (Stanton et al., 2022). There are several reasons why Zn deficiency may occur in soil, including low soil pH, high soil phosphorus levels, and sandy or coarse soils with low organic matter content (López-Rayo, 2023). When plants are deficient in Zn they may exhibit several symptoms, such as stunted growth, yellowing of leaves, and delayed maturity. Additionally, plants experiencing Zn deficiency may exhibit higher levels of reactive oxygen species (ROS), reduced root growth, and weakened stems. As a result, they become more susceptible to damage from pests and diseases (Zeng et al., 2021). Zn plays a critical role in several essential plant metabolic processes, including photosynthesis, hormone regulation, and enzyme activation (Lay-Pruitt et al., 2022; Akhtar et al., 2019). When a plant is deficient in Zn, it can result in

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significant metabolic disturbances that impact plant growth and development (Assunção, 2022). One of the primary effects of Zn deficiency is a reduction in chlorophyll synthesis, which can lead to decreased photosynthesis rates and lower plant productivity (Zhang et al., 2019). Furthermore, Zn is a cofactor for several enzymes involved in essential metabolic processes, such as carbohydrate metabolism and protein synthesis (Mallikarjuna et al., 2020). Therefore, Zn deficiency can lead to stunted growth and delayed maturity in plants.

Plants have developed several mechanisms to cope with Zn deficiency and maintain their metabolic functions. One of the primary strategies used by plants is to increase their Zn uptake from the soil. Plants can release organic acids, such as citric acid, from their roots, which can help solubilize Zn in the soil and make it more available for uptake (Saleem et al., 2022; Widodo et al., 2010). Additionally, plants can produce specialized transporters, called ZIP transporters, which can enhance the uptake of Zn ions from the soil (Krishna et al., 2020; Claus and Chavarría-Krauser, 2012). Another strategy used by plants is to reallocate Zn from older leaves to newer ones through the phloem tissue, ensuring that critical metabolic functions, such as photosynthesis, continue to occur (Stomph et al., 2014). A number of transporters have been identified that are involved in Zn and Fe accumulation including but not limited to Fe-regulated transporter-like proteins (ZIPs), natural resistance-associated macrophage protein (NRAMPs), and metal tolerance proteins (MTPs) in plants (Bashir et al., 2019; Cai et al., 2019). ZIP transporters play a crucial role in Zn homeostasis and are highly regulated under different Zn availability conditions and the genetic background of the plant species.

Microbes, such as bacteria and fungi, can play a critical role in Zn uptake in plants by enhancing the availability of this micronutrient in the soil (González-Guerrero et al., 2016; Housh et al., 2021). These microbes can solubilize Zn in the soil, making it more accessible for plant uptake. Some soil bacteria and fungi can produce organic acids, such as gluconic acid and citric acid, which can help dissolve Zn compounds in the soil, releasing free Zn ions that are more readily taken up by plant roots (Costerousse et al., 2017; Jain et al., 2020). Additionally, some microbes can produce specialized enzymes, such as phosphatases, that can break down organic phosphorus compounds in the soil, reducing competition between phosphorus and Zn for plant uptake (Kisko et al., 2018; Singh and Satyanarayana, 2011). Recently, arbuscular mycorrhizal fungi (AMF) that form symbiotic relationships with plant roots are reported to enhance the uptake of Zn and other nutrients (Wang et al., 2022; Saboor et al., 2021). These fungi can produce hyphae that penetrate the root cells, increasing the surface area of the roots and allowing for greater nutrient uptake (Feng et al., 2022). Studies have shown that AM fungi can enhance Zn uptake in plants by increasing the surface area of the roots, producing specialized transporters that facilitate Zn uptake, and producing organic acids that help solubilize Zn in the soil (Saboor et al., 2021). Furthermore, AM fungi increased the expression of several genes involved in Zn uptake and transport, further supporting the role of these fungi in plant Zn uptake (Cardini et al., 2021; Coccina et al., 2019).

Broccoli (*Brassica oleracea* var. oleracea) is a widely recognized vegetable abundant in vitamin C, fiber, and carotenes (Sikora and Bodziarczyk, 2012). Broccoli is a non-mycorrhizal plant species that typically occurs in habitats where competition with other plants is low, and there is a high availability of soil phosphorus (Olsson and Tyler, 2004). Since Zn deficiency is a limiting factor for vegetable crops, growers typically use soil amendments and fertilizers that are not only costly but also environmentally unsafe. To address this challenge, the development of beneficial microbiomes offers a sustainable alternative for inducing abiotic stress tolerance in plants. Despite being ranked among the top ten economically significant vegetables, other crops of the Brassica genus, including broccoli, have not garnered substantial focus concerning microbiome-assisted enhancement (Francisco et al., 2017). Specifically, the beneficial role of AMF in response to Zn deficiency and its effects on physiological responses and Zn induction in broccoli are largely unknown. Therefore, the present study aimed to determine whether AMF mitigates Zn deficiency in broccoli via beneficial symbiotic interactions. This study also intends to investigate the role of metabolites, Zn-related genes, and antioxidant systems in Zn-deficient broccoli inoculated with AMF.

2. Materials and methods

2.1. Plant cultivation and AMF treatment

Broccoli seeds were subjected to surface sterilization using 70 % ethanol for a duration of 5 min before being germinated on tissue paper in a germination tray at room temperature, with distilled water as a supplement. The seeds were allowed to germinate on moist tissue paper, which was kept damp with deionized water, in a dark environment at room temperature for 3 d. Only healthy and uniform seedlings were selected for transplantation. Vermiculite supplemented with basal half-strength nutrient solution (Hoagland and Arnon, 1950) was used with the following nutrient concentrations (μM): KNO_3 (16000), $Ca(NO)_3.4H_2O$ (6000), $NH_4H_2PO_4$ (4000). $MgSO_4.7H_2O$ (2000), KCI (50), H_3BO_3 (25), Fe-EDTA (25), $MnSO_4.7H_2O$ (2). $Na_2MoO_4.2H_2O$ (0.5), $CuSO_4.5H_2O$ (0.5), and Zn as $ZnSO_4$ at two levels viz. 0.1 μM (Zn-deficient) and 2.0 mM (Zn sufficient/control). The AMF inoculum mixture consisted of endomycorrhizal spores from three species: *Rhizophagus irregularis, Rhizophagus aggregatum*, and *Claroideoglomus etunicatum* (Organic Gardening, New South Wales, Australia). There were four treatments: control (2.0 mM $ZnSO_4$); -Zn (0.1 μM $ZnSO_4$); -Zn + AMF (0.1 μM $ZnSO_4$ + 0.2 g AMF spore) and control with AMF (2.0 mM $ZnSO_4$ + 0.2 g AMF spore). The pH of the medium was adjusted to 6.0 using KOH (1M). Both unstressed control plants and plants under Zn deficiency, with or without AMF, were grown simultaneously, each having at least three replicates, using a completely randomized design. The plants were cultivated in a growth chamber under controlled conditions, with a temperature of 25°C, 60 % relative humidity, and a light intensity of 200 $\mu mol^{-2}s^{-1}$. The light conditions followed a long-day pattern of 14 h of light followed by 10 h of darkness. After transferring the plants to the vermiculite, they were allowed to grow for a period of 3 weeks before harvesting.

2.2. Morphological growth parameters

Various morphological growth parameters were assessed on plants cultivated in a solution culture. The parameters measured included shoot height, root length, shoot fresh weight, and root fresh weight. To measure the total roots developed by each plant sample, the roots were carefully washed in distilled water to eliminate any residual elements and then promptly blotted dry with tissue paper. The fresh weight of the root and shoot was directly measured using an electronic balance after the plants were harvested. Shoot height and root length were measured using a centimeter measuring scale. After the measurements, the roots and leaves of the plants were manually separated and then dried in an oven at a temperature of 80 °C for 3 d to remove all moisture before measuring the dry weight.

2.3. Determination of chlorophyll score and photosynthetic parameters

The chlorophyll content of young, fully expanded leaves was assessed using a chlorophyll meter (atLEAF, United States). Additionally, the chlorophyll fluorescence characteristics, namely Fv/Fm (quantum efficiency of photosystem II) and Pi_ABS (photosynthesis performance index), of these leaves were measured using a Fluor Pen (FP) 100 portable equipment (Photon Systems Instruments, Czech Republic). Before the measurements, the young, expanded leaves were acclimated in darkness for 1 h.

2.4. Determination of mycorrhizal colonization

The mycorrhizal colonization was visualized in roots cut into 1-2 cm. Then root samples were initially submerged with 5 % (w/v) solution KOH at 60 °C for 8 min. The root samples were then cleaned, submerged with 1 % HCl solution, and incubated for 5 min. Afterward, 0.05 % trypan blue was added to the samples and kept at 60 °C for 10 min. To absorb excess stain, these samples were kept on tissue paper and then transferred into Petri dishes with lactoglycerol solution for 40 min. The root section was mounted on the slide with lactoglycerol followed by visualization in a digital microscope at 40 × magnification. Furthermore, we calculated the colonization efficiency of AMF in roots Islam et al. (2022)

2.5. Analysis of stress indicators

Cell death was assessed using a modified Evans blue method (Zhao et al., 2005). Initially, the plant root and shoot were exposed to a 0.25 % Evans blue emulsion for approximately 15 min. Subsequently, the suspension was treated with 1 ml of 80 % ethyl alcohol for 10 min at room temperature. The samples were then incubated at 50 °C for 15 min in a water bath, followed by rotation at 12,000 rpm for 10 min. The optical absorption of the supernatant was then measured at 600 nm using a spectrophotometer (Shimadzu, Japan). The cell death (%) was calculated on the basis of the fresh weight of each sample as previously described (Zhao et al., 2005).

To evaluate membrane integrity, electrolyte leakage was utilized as a measurement (Lutts et al., 1996). This assessment was conducted separately for roots and shoots using an electrical conductivity meter. Briefly, the root and shoot samples were washed with distilled water and then cut into 1 cm segments. Afterward, each segment was placed in an individual vial containing 20 ml of deionized water. These vials were then incubated at room temperature (25 °C) on a shaker, with a speed of 100 rpm, for 2 h. Following the incubation period, the electrical conductivity (EC) of the bathing solution was measured, serving as an indicator of the amount of electrolyte leakage that occurred during the incubation process.

To analyze the levels of hydrogen peroxide (H_2O_2), both roots and shoots were mixed with 0.1 % trichloroacetic acid (Alexiva et al., 2001). The mixture was then subjected to centrifugation at 10,000 rpm for 15 min to separate the cellular debris from the extracted fluid. The upper aqueous phase obtained after centrifugation was mixed with potassium iodide (1 M) and phosphate buffer (10 mM) with a pH adjusted to 7.0. This mixture was kept at dark for 60 min to allow for a chemical reaction. Finally, the distillate obtained from the reaction was measured using a Shimadzu UV-1650PC spectrophotometer at an optical frequency of 390 nm.

2.6. Determination of Zn and Fe in root and shoot

After harvesting, broccoli root samples were carefully washed three times with deionized water to eliminate any surface-associated arbuscular mycorrhizal fungi (AMF). Subsequently, the roots were subjected to incubation at 4 °C in two solutions: the first solution containing 10 mM MES (2-(N-morpholino)ethanesulfonic acid) and the second solution comprising 10 mM MES and 1 mM EDTA (ethylene-diamine-tetra-acetic acid). Following the incubation, the roots were washed 2-3 times with deionized water. Both the broccoli root and shoot samples were then individually dried in an oven at 80 °C for 72 h. After the drying process, the samples were weighed, and subsequently, they were dissolved using a solution of $HClO_4/HNO_3$ (1:3). The concentrations of Zn and Fe in the digested solution were determined using ICP-MS (inductively coupled plasma mass spectrometry) by comparing them with standard known solutions of these (Agilent 7700, USA).

2.7. Analysis of antioxidant enzyme

Plant tissues were homogenized in 1.5 ml K-phosphate buffer. The mixture was then centrifuged at 15,000 rpm for 20 min and the supernatant was separated in a new tube. To determine catalase (CAT), the plant extracts (100 µL) were mixed with 100 mM KH_2PO_4 800 µL (pH 7.0) and 200 mM H_2O_2 400 µL. The POD activity was tested by an assay solution mixed with 1 ml of 0.05 M pyrogallol solution, 400 µl of 200 mM H_2O_2, and 100 µl extract. The change in absorbance was recorded from 30 s up to 1.5 min in a spectrophotometer (Shimadzu, Japan). One unit of peroxidase is defined as the change in absorbance/minute at 430 nm. The POD activity was calculated using the extinction coefficient for purpurogallin ($\varepsilon = 12\ mM^{-1}cm^{-1}$). To measure the activity of ascorbate peroxidase (APX), the plant extracts were mixed with a solution containing 0.1 mmol EDTA, 50 mmol KP-buffer (potassium phosphate buffer) with a pH of 7.0, 0.1 mmol H_2O_2, 0.5 mmol ascorbic acid, and 0.1 ml of the extraction. The APX activity was then calculated using an extinction coefficient based on the optical density recorded at 290 nm (Goud and Kachole, 2012). For determining superoxide dismutase (SOD) activity, the plant extracts (100 µL) were mixed with a solution containing 0.1 mmol EDTA, 50 mmol $NaHCO_3$ with a pH of 9.8, and 0.6 mmol epinephrine. After 4 min, the formation of adrenochrome was measured at 475 nm to quantify the SOD activity (Goud and Kachole, 2012).

2.8. Determination of S-metabolites

A high-performance liquid chromatography (HPLC) with a dual Waters 2489 detector was used to analyze the concentration of cysteine (Cys), glutathione (GSH), and methionine (Met) in the root using the wavelengths of 280 nm and 360 nm. The HPLC system was equipped with a C18 reverse-phase column, and a gradient condition was employed using 100 % acetonitrile as the mobile phase (Kabir et al., 2016; Lindberg et al., 2007). The mobile phase consisted of buffer A (0.1 % TFA in water) and buffer B (0.1 % TFA in 80 % acetonitrile) with the following gradient: 1–24 min, 100 % A; 25–34 min, 100 % B; and 35–40 min, 100 % A. The standard of targeted metabolites (Cys, GSH and Met) and samples were diluted 100 times and filtered through 0.22 µm Minisart Syringe filters before injecting into the HPLC system to ensure accurate and precise analysis.

2.9. Expression of genes relative to Zn uptake and transport in roots

We have selected a few candidate genes, such as *BoZIP1* (ZRT/IRT-like protein 1 gene), BoNRAMP1 (natural resistance-associated macrophage protein 1) and BoMTP1 (metal tolerance protein 1), which are responsible for Zn uptake and transport in plants. The total RNA in the roots of plants was extracted using a mortar and pestle and liquid nitrogen as instructed by the SV total RNA isolation system (Promega, United States). The final yield and purity of the isolated RNA were then measured using a Nanodrop Spectrophotometer. The first-strand cDNA was synthesized from RNA using the cDNA reverse transcription kit (Promega, United States). The real-time PCR was performed in Eco™ real-time PCR system (Illumina, USA) using gene-specific primers (Supplementary Table S1) designed based on the Arabidopsis gene database. The PCR program was as follows: 95 °C for 30 s, followed by 40 cycles at 95 °C for 5 s, 60 °C for 30 s. The expression level of target genes was calculated using GAPDH as an internal control.

2.10. Statistical analysis

The experiments were conducted using a completely randomized design (CRD) with three independent biological replications for each sample. To determine the significance of the data within individual groups, analysis of variance (ANOVA), and post-hoc comparisons were performed using Duncan's Multiple Range Test (DMRT). The statistical

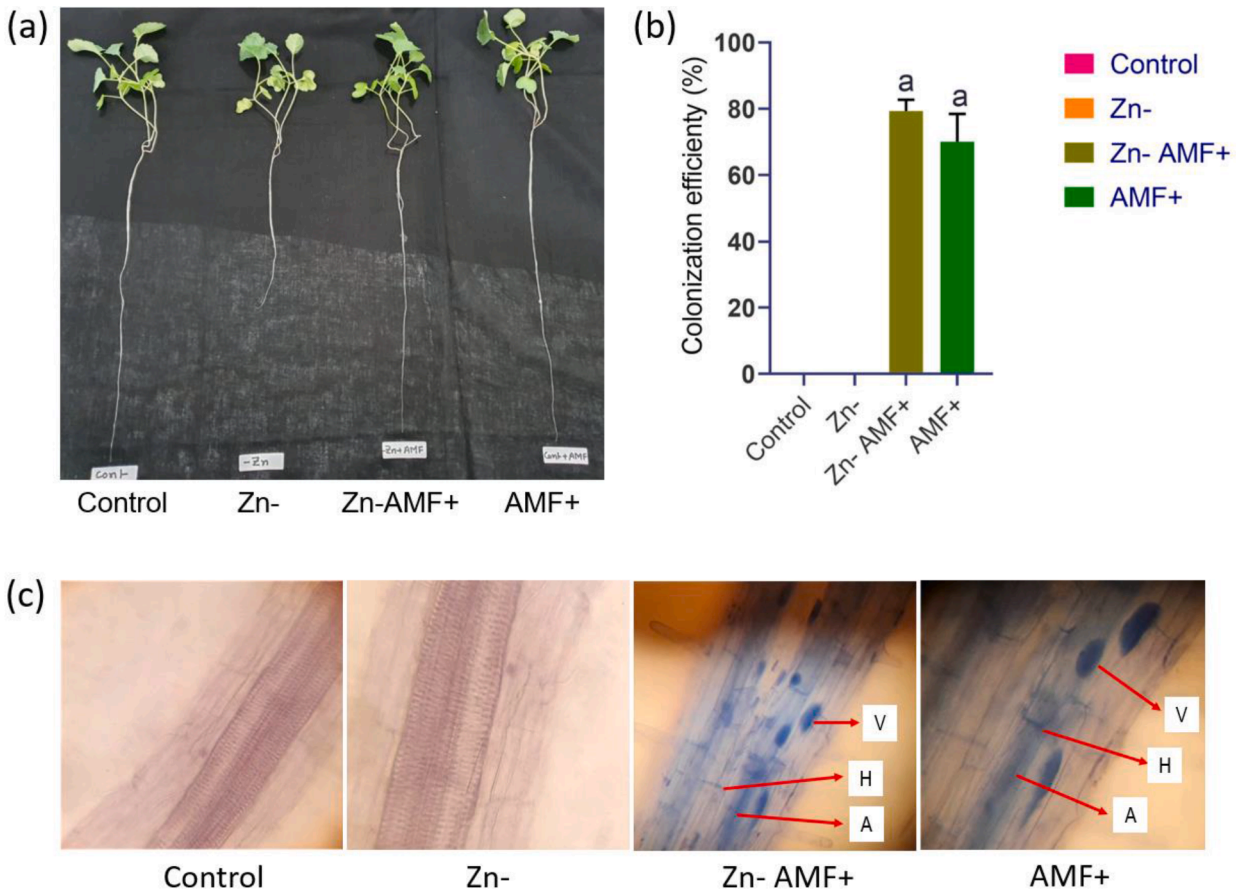


Fig. 1. The effect of AMF on broccoli plants cultivated for 21 d in different growth conditions (control, Zn-, Zn- AMF+, AMF+): plant phenotype (a), colonization efficiency in roots (b), AMF structure in roots: hyphae (H), arbuscule (A) and vesicle (V). The broccoli roots were visualized with a light microscope.

analysis was carried out using the SPSS Statistics 20 system. Furthermore, Graph Pad Prism 6 software was used to create graphical figures to visually represent the results of the experiments.

3. Results

3.1. Establishment of AMF colonization

In this study, plant health was significantly improved due to the plant-AMF association under Zn deficiency (Fig. 1b). Although we did not see any AMF colonization efficiency in the roots of broccoli cultivated under control and Zn-deficient conditions, the addition of AMF (Zn- TH+ and AMF+) showed 60–80 % colonization efficiency (Fig. 1b). The roots of broccoli plants supplemented with AMF displayed visible hyphae, arbuscule and vesicle. On the other hand, the roots of plants without AMF supplemented did not display any AMF structure (Fig. 1c).

3.2. Improvement in morphological features

The root length and shoot height were measured in broccoli grown in four treatments (control, -Zn, -Zn + AMF and control + AMF). The root length of broccoli was significantly decreased under Zn deficiency compared to controls. However, the root length was significantly increased under Zn deficiency with AMF compared to Zn-deficient plants. Plants cultivated only with AMF showed similar root lengths to those plants cultivated under control conditions (Fig. 2a). The shoot height of broccoli was significantly decreased under Zn deficiency relative to controls. The shoot height was significantly increased in Zn-deficient plants inoculated with AMF compared to Zn-starved plants. Plants cultivated only with AMF showed similar shoot height to those plants cultivated under Zn-sufficiency (Fig. 2b). Furthermore, the root dry weight of broccoli was significantly decreased under Zn deficiency in comparison with Zn-sufficient controls. However, the addition of AMF caused a significant increase in root dry weight of Zn-deficient plants compared to Zn-deficient plants. Plants cultivated only with AMF showed similar root dry weight to those plants cultivated under control conditions (Fig. 2c). Also, the shoot dry weight was significantly decreased due to Zn deficiency relative to controls. However, the shoot dry weight showed a significant increase in Zn-deficient plants inoculated with AMF relative to plants cultivated solely with Zn-deficiency. Plants cultivated only with AMF showed similar shoot dry weight to those plants cultivated under Zn-sufficient controls (Fig. 2d).

3.3. Photosynthetic parameters

Several leaf parameters, such as SPAD (the soil plant analysis development), Fv/Fm (quantum efficiency of photosystem II), and Pi_ABS (photosynthesis performance index) were measured to determine the effect of AMF on preventing photosynthesis damage caused by Zn deficiency in broccoli. In the absence of sufficient Zn, the SPAD, Fv/Fm, and Pi_ABS values exhibited a significant decrease compared to the control group (Fig. 3a-c). However, when Zn-deficient plants were inoculated with AMF, there was a significant improvement in these leaf features compared to plants that lacked AMF under Zn deficiency. Additionally, plants grown with AMF and sufficient Zn demonstrated similar leaf chlorophyll levels and photosynthetic parameters as the control group and plants inoculated with AMF under Zn deficiency (Fig. 3a-c).

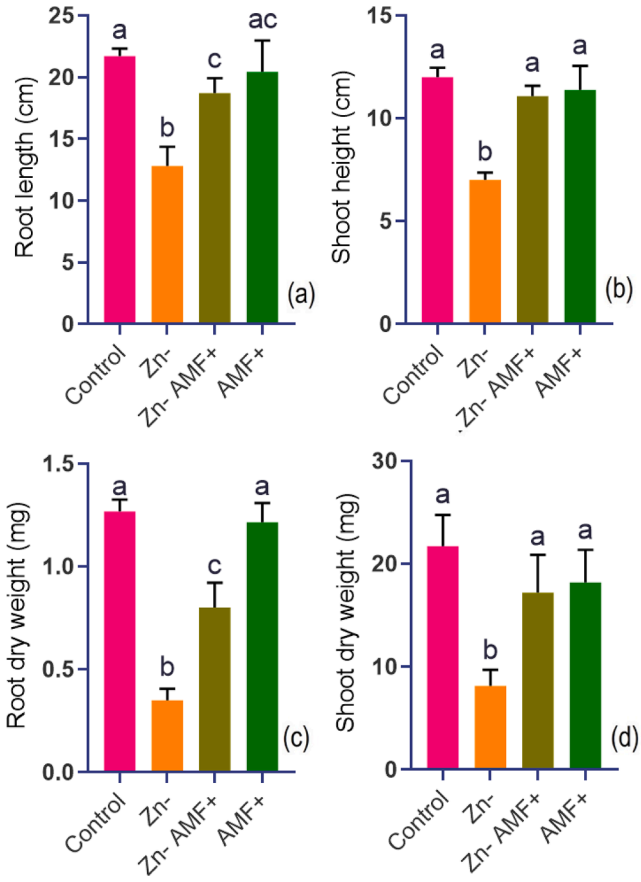


Fig. 2. The effect of AMF in morphological features of broccoli plants cultivated in different growth conditions of Zn availability and AMF inoculation: root length (a), shoot height (b), root dry weight (c) and shoot dry weight (d). Different letters indicate significant differences between means ± SD of treatments (n = 3) at $p \leq 0.05$ level.

3.4. Changes in cell abiotic stress indicators

The cell death (%) in the roots of broccoli was significantly increased under Zn deficiency relative to controls. However, the addition of AMF on Zn-deficient plants showed a significant decrease in cell death (%) in contrast to Zn-deficient broccoli. Plants cultivated solely with AMF showed a similar cell death (%) in the roots to those plants cultivated under control conditions (Fig. 4a). The cell death (%) was significantly increased in the shoot of broccoli plants with Zn deficiency relative to controls. However, the Zn-deficient plants inoculated with AMF showed a significant decline in cell death (%) in the shoot relative to Zn-deficient plants, which is similar to the controls and plants cultivated solely with AMF (Fig. 4a). In addition, the electrolyte leakage in roots and shoots was significantly increased under Zn deficiency compared to controls (Fig. 4b). The Zn-starved plants inoculated with AMF showed a significant decrease in electrolyte leakage in both root and shoot relative to Zn-deficient plants. Plants cultivated only with AMF showed similar electrolyte leakage in roots and shoots to those plants cultivated under control and plants inoculated with AMF under Zn-shortage (Fig. 4b). Lastly, H_2O_2 in roots and shoots of broccoli significantly increased in Zn-deficient plants relative to Zn-sufficient controls (Fig. 4c); these were significantly decreased while Zn-starved plants were treated with AMF relative to plants solely cultivated under Zn-deficiency. Plants cultivated only with AMF showed similar (H_2O_2) in roots and shoots to those plants cultivated under controls and Zn-AMF+ conditions (Fig. 4c).

3.5. Zn and Fe concentrations in roots and shoots

The Zn concentrations in the root and shoot of broccoli significantly decreased under Zn deficiency compared to Zn-sufficient controls (Fig. 5a). When Zn-deficient plants were inoculated with AMF, the concentration of Zn in both tissues showed a significant increase relative to plants solely cultivated under Zn-deficiency. Plants solely cultivated under AMF with sufficient Zn showed a significant increase of Zn in both root and shoot compared to controls and plants cultivated under Zn-deficient conditions with or without AMF inoculation (Fig. 5a). Also, Fe concentrations in roots and shoots of broccoli were significantly decreased under Zn deficiency relative to controls; however, the addition of AMF under Zn-deficiency caused a significant increase in Fe levels in contrast to Zn-deficient broccoli (Fig. 5b). Plants cultivated only with AMF showed similar Fe concentrations in roots and shoots to those plants cultivated under control conditions (Fig. 5b).

3.6. Changes in antioxidant enzymes

The CAT activity in the roots of broccoli was significantly increased under Zn deficiency relative to Zn-sufficient controls. The addition of AMF with or without Zn-deficiency showed no significant changes in CAT activity in the roots of plants compared to Zn-deficient plants. The CAT activity in plants inoculated with AMF showed a similar trend to that of controls and Zn-deficient plants inoculated with AMF (Fig. 6a). The CAT activity showed no significant variations in the shoot of plants in the presence or absence of Zn-deficiency or AMF (Fig. 6a). The POD activity in roots of broccoli showed no significant changes due to Zn-

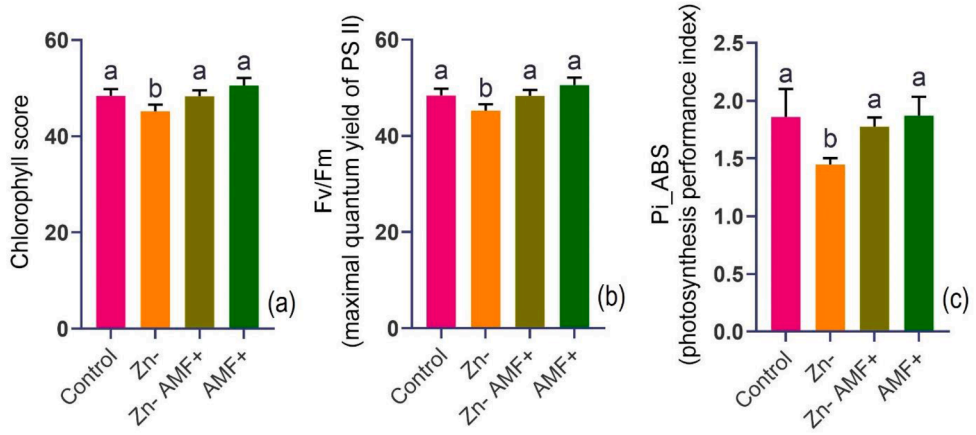


Fig. 3. The effect of AMF in leaf photosynthetic features of broccoli plants cultivated in different growth conditions of Zn availability and AMF inoculation: chlorophyll score (a), Fv/Fm (b), and Pi_ABS (c). Different letters indicate significant differences between means ± SD of treatments (n = 3) at $p \leq 0.05$ level.

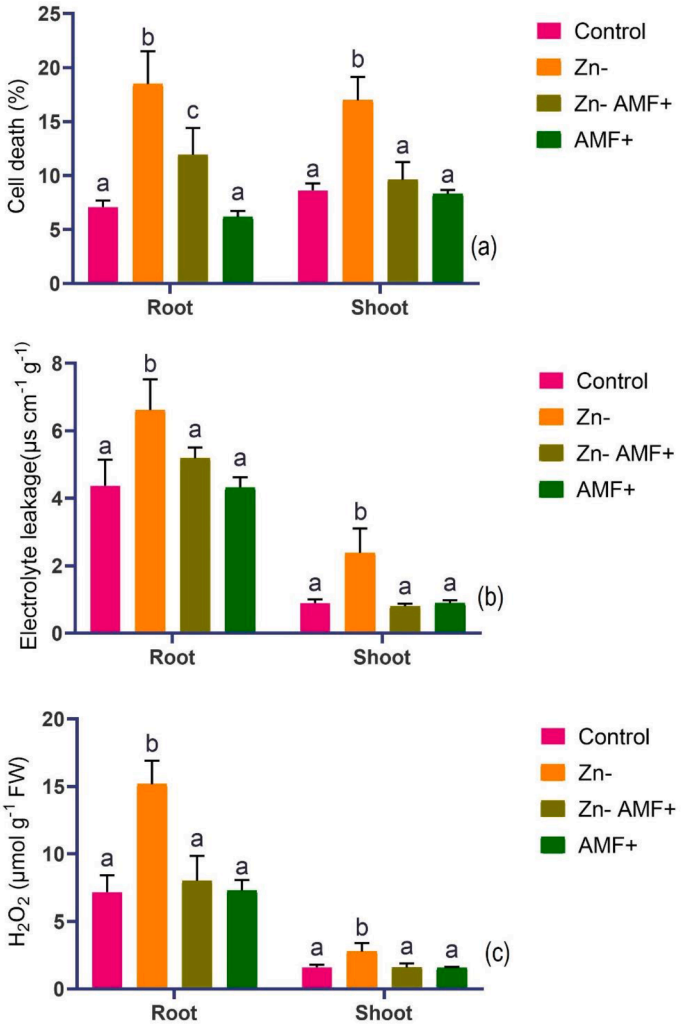


Fig. 4. The effect of AMF in stress indicators of broccoli plants cultivated in different growth conditions of Zn availability and AMF inoculation: cell death % (a), electrolyte leakage (b) and H_2O_2 (c) in root and shoot. Different letters indicate significant differences between means ± SD of treatments (n = 3) at $p \leq 0.05$ level.

deficiency relative to controls but significantly induced in AMF-inoculated Zn-deficient plants in comparison with Zn-deprived plants (Fig. 6b). Plants cultivated only with AMF showed similar POD activity in roots to those of plants cultivated under control conditions. Furthermore, the POD activity in the shoot showed no significant changes regardless of the presence or absence of Zn-deficiency or AMF (Fig. 6b). The APX activity in roots of broccoli was significantly increased under Zn deficiency compared to controls (Fig. 6c). When AMF was added in Zn-deficient plants, the APX activity showed a significant increase relative to Zn-deficient plants. Plants cultivated only with AMF showed similar APX activity in roots to those plants cultivated under control conditions (Fig. 6c). Also, the APX activity in the shoot was significantly induced due to Zn-deficiency with or without AMF inoculation compared to controls (Fig. 6c). However, the APX activity in the shoot showed a similar pattern when plants were cultivated under control conditions with AMF inoculation (Fig. 6c). Lastly, the SOD activity showed no significant changes in the roots of Zn-deficient plants relative to controls (Fig. 6d). The SOD activity significantly induced in the roots of Zn-deficient plants inoculated with AMF compared to Zn-starved plants. Plants cultivated solely with AMF showed similar SOD activity in the roots similar to the plants cultivated under control conditions (Fig. 6d). The SOD activity showed no significant changes in the shoot of plants exposed to Zn-deficiency inoculated with or without AMF (Fig. 6d).

3.7. Changes of S-metabolites in the roots

The Cys concentration in roots and shoots of broccoli was significantly decreased under Zn deficiency compared to controls (Fig. 7a). However, Cys levels in roots and shoots were significantly increased under Zn deficiency inoculated with AMF compared to Zn-deficient plants. Plants cultivated only with AMF showed a similar Cys concentration in both tissues to those plants cultivated with Zn-sufficiency (Fig. 7a). Under Zn shortage, the GSH concentration in broccoli roots was considerably lower than in controls (Fig. 7b). When Zn-deficient plants were inoculated with AMF, the roots showed considerably higher GSH levels than Zn-deficient plants. The GSH levels in the roots of plants grown exclusively with AMF were comparable to those of plants cultivated with sufficient Zn (Fig. 7b). Plants showed no significant changes in GSH levels in the shoot among different treatments of Zn-deficiency and AMF (Fig. 7b). The Met showed no significant changes either in root and shoot in the presence or absence of Zn-deficiency and AMF (Fig. 7c).

3.8. Changes of candidate genes in the roots related to Zn uptake

The qPCR results showed a significant upregulated in the expression of the *BoZIP1* gene in roots of Zn-deficient broccoli relative to controls (Fig. 7d). Furthermore, the expression of *BoZIP1* significantly induced in Zn-deficient plants inoculated with AMF in contrast to Zn-deficient

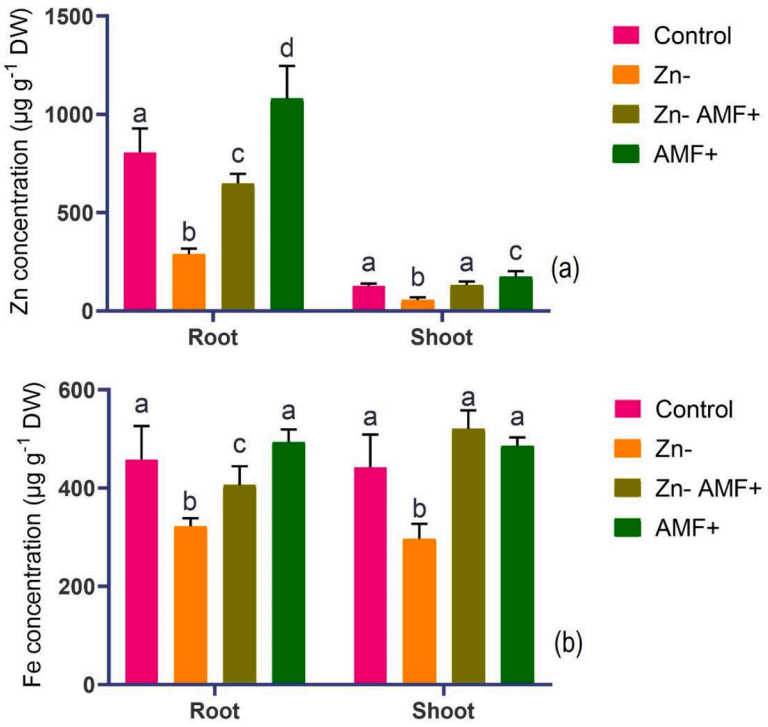


Fig. 5. Changes in Zn (a) and Fe (b) concentration in root and shoot of broccoli plants cultivated in different growth conditions of Zn availability and AMF inoculation. Different letters indicate significant differences between means ± SD of treatments (n = 3) at $p \leq 0.05$ level.

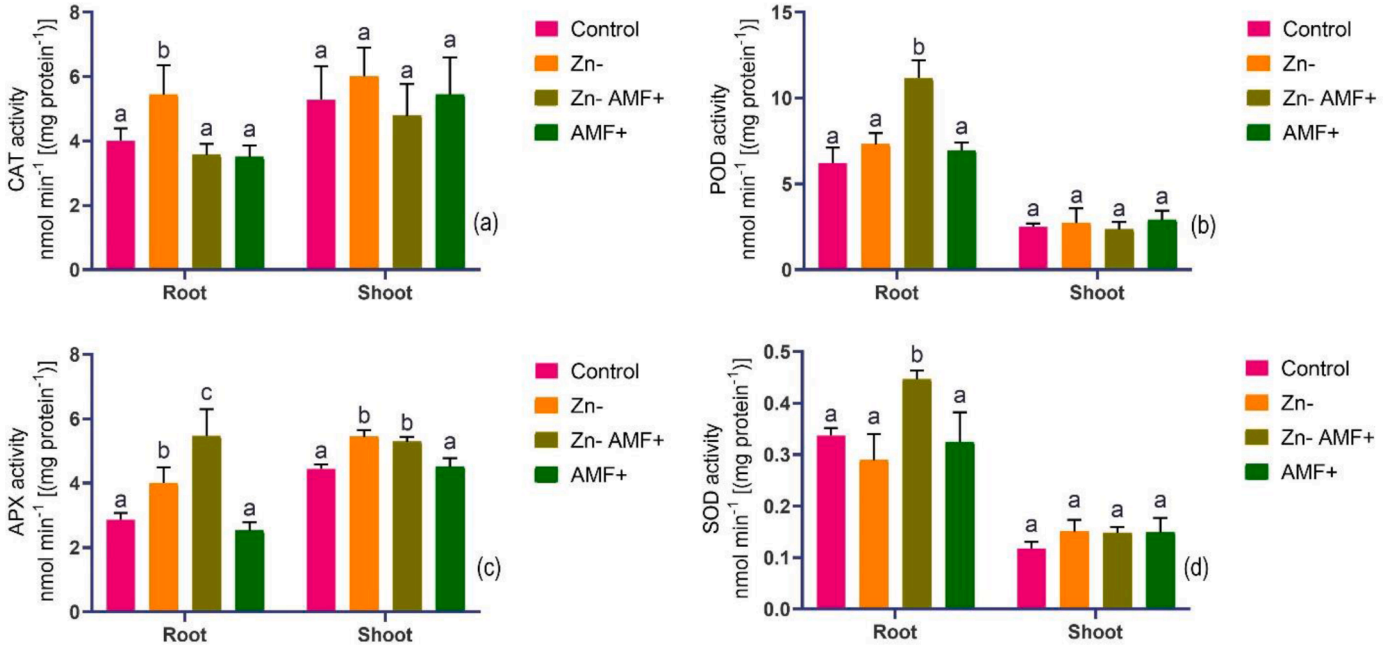


Fig. 6. The effect of AMF in antioxidant enzymes of broccoli plants cultivated in different growth conditions of Zn availability and AMF inoculation: CAT(a), POD (b), APX (c) and SOD (d) activities in root and shoot. Different letters indicate significant differences between means ± SD of treatments (n = 3) at $p \leq 0.05$ level.

plants. The Zn-sufficient plants cultivated solely with AMF showed significant induction of *BoZIP1* in roots compared to controls but were significantly higher in expression compared to Zn-deficient plants (Fig. 7d). Another Zn-transporter *BoNRAMP1* showed no changes in the roots of Zn-deficient broccoli relative to controls. However, the Zn-deprived plants inoculated with AMF showed a significant upregulation of *BoNRAMP1* in roots than Zn-deficient plants. Zn-sufficient plants inoculated with AMF showed significant downregulation of *BoNRAMP1* in the roots of Zn-deficient plants cultivated with AMF, but the expression was significantly higher than those of controls and Zn-deficient plants (Fig. 7d). The Zn-deficient plants cultivated with or without AMF showed a significant decline in the expression of *BoMTP1* in the roots compared to controls. Zn-sufficient plants solely cultivated with AMF showed similar expression of *BoMTP1* in roots relative to controls (Fig. 7d).

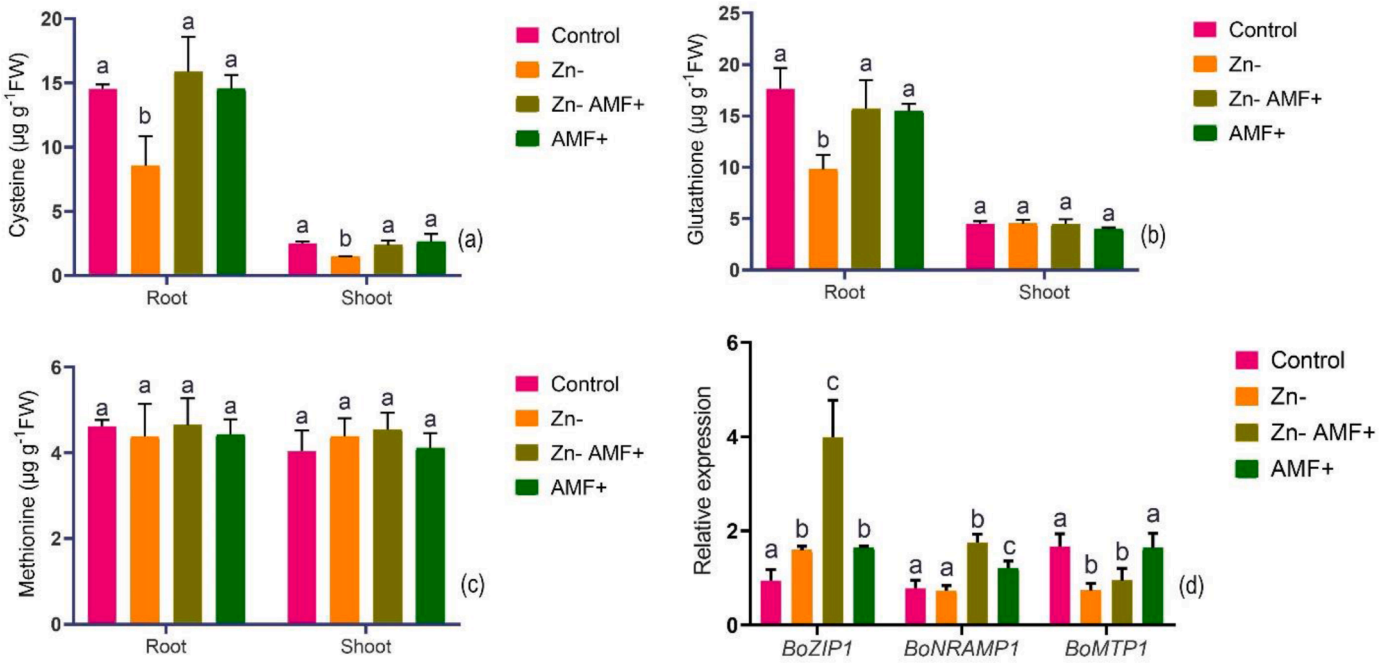


Fig. 7. Changes in S-metabolites in root and shoot as well as expression of genes (*BoZIP1, BoNRAMP1* and *BoMTP1*) in roots of broccoli plants cultivated in different growth conditions of Zn availability and AMF inoculation: cysteine (a), glutathione (b), methionine (c) and relative expression (d). Different letters indicate significant differences between means ± SD of treatments (n = 3) at $p \leq 0.05$ level.

4. Discussion

Elimination of nutrient deficiency using effective microorganisms is an eco-friendly approach for sustainable nutrient security with smart vegetable production. In this study, Zn deficiency induced chlorosis, inhibited plant growth, and development, as well as disturbed the redox status in plants. However, the supplementation of AMF significantly reversed these attributes in broccoli plants deficient with Zn. The colonization of AMF in non-host plants like broccoli is unlikely, but this study encourages future efforts to study this phenomenon. Additionally, broccoli plants possess the ability to utilize AMF-mediated benefits from the attached communities capable of hosting AMF. Furthermore, this study not only unveils the symbiosis of AMF in diminishing Zn deficiency but also reveals the benefit of AMF in improving morphological, physiological, and biochemical traits in broccoli.

4.1. AMF colonization in non-host broccoli

In this study, AMF inoculation showed visible hyphae in the roots which is in agreement with the findings reported by another non-host mycorrhizal species *Arabidopsis thaliana* (Veiga et al., 2013). However, it is known that mycorrhizal colonization did not occur within the roots of broccoli but introducing AMF inoculation intercropped with sesame led to a systemic elevation of indole glucosinolates in both the roots and leaves of broccoli (Tong et al., 2015). Fungal hyphae penetration, devoid of arbuscule formation, has been noted on multiple occasions (Allen et al., 1989; Rinaudo et al., 2010). Some studies suggest that certain chemical signals released by non-host plants can attract AMF (Wang et al., 2021). Therefore, it is possible that while broccoli plants may not be the preferred hosts, the chemical signals emitted by their roots might still be recognized by the fungi, leading to colonization. By definition, the roots of a non-host plant are never colonized, but this determination relies on the evidence of absence. An example is *Buddleja davidii*, initially classified as a non-host plant but later revealed to be a genuine AMF host (Dickie et al., 2007). Instead, the non-host status of a species may still be assigned in cases where AMF colonization occurs but does not align with a recognizable functional type (Brundrett and Tedersoo, 2018). Furthermore, it is reported that *Trichoderma harzianum* promotes the accessibility of arbuscular mycorrhizal fungi to the roots of non-host Brassicaceae and enhances overall plant productivity (Poveda et al., 2019). Although we used vermiculite for plant cultivation in the greenhouse, the ineffective sterilization process may also lead to the presence of other biological communities capable of hosting the AMF attached to the broccoli roots. Therefore, our studies suggest that the interactions between non-mycorrhizal species and AMF seem to lean towards facilitative rather than strongly competitive under stressed conditions. However, further efforts are needed to elucidate the underlying mechanisms of AMF-broccoli symbiosis.

4.2. AMF improved plant biomass and photosynthetic attributes

In this study, Zn deficiency caused reduced root and shoot biomass in broccoli. In this study, plants cultivated in vermiculite supplemented with a nutrient solution adjusted to pH 6.0 were found to be promising for studying AMF colonization, as evidenced by microscopic visualization of AMF structures in broccoli roots. Vermiculite-based plant cultivation provides an excellent platform for microbiome studies, offering a balanced and supportive environment for plant-microbe interactions (Ferrari et al., 2020). The ideal pH for plant growth falls within the range of 5.5 to 6.5, ensuring optimal nutrient availability. This range is also conducive to the thriving of most soil microbes, partly due to the flourishing growth of plants within this pH range (Zhou et al., 2016). Conversely, fungal activities thrive under somewhat acidic pH conditions, elucidating their dominance in acidic soils within forests as opposed to range land soils, sub-humid areas, and arid prairies. These latter environments, being mildly acidic, are primarily dominated by bacteria (Zifcakova, 2020). We found that the supplementation of AMF considerably improved plant length and dry biomass in broccoli under Zn deficiency. In addition, chlorophyll pigment and photosynthetic attributes remarkably improved in broccoli leaves after AMF supplementation even though broccoli plants were cultivated under Zn deficient conditions. These findings suggest that AMF is capable of restoring Zn deficiency and improving plant traits in broccoli. Zn deficiency causes growth retardation, and leaf chlorosis, which disrupts the regulation of

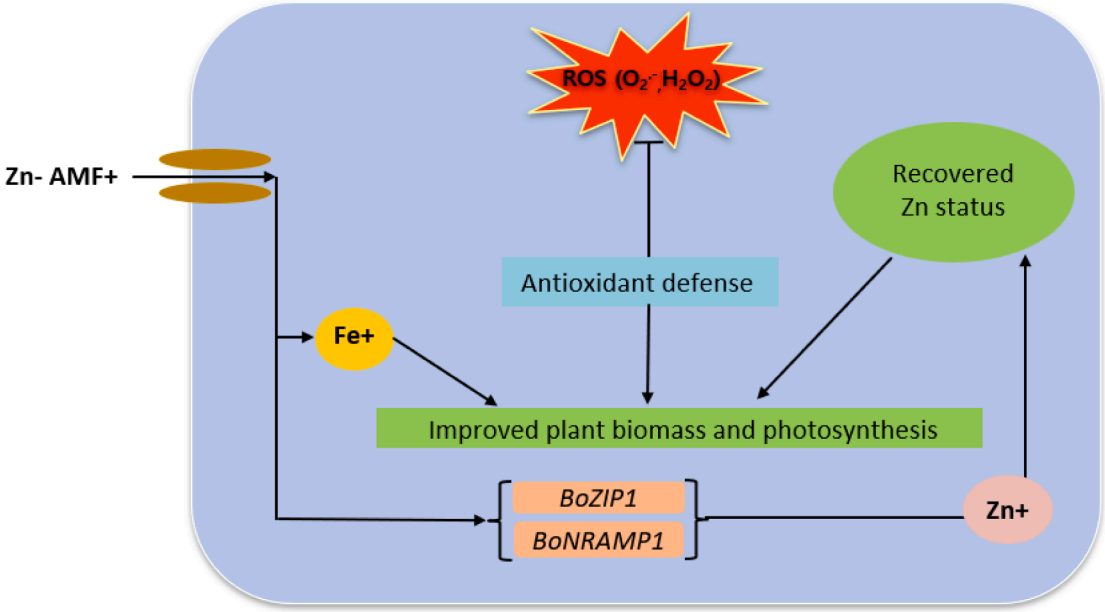


Fig. 8. Mechanistic insights of AMF-mediated Zn deficiency alleviation in broccoli. The supplementation of AMF in Zn-deficient conditions induces Fe concentration and the expression of *BoZIP1* and *BoNRAMP1* which leads to improved Zn levels in recovering Zn status. Zn deficiency induces ROS ($O_2^{\cdot-}$, H_2O_2) in plant cells, while the addition of AMF promotes the antioxidant defense to cope with the Zn-deficiency-induced oxidative stress. The processes involving AMF-mediated symbiosis improve plant biomass and photosynthesis in Zn-starved broccoli.

plant metabolism (Hamzah et al., 2022). Zn deficiency changes the declined chloroplast structure that influences the photosynthesis and PSII efficiency in plants (Zhang et al., 2019; Zhao et al., 2017), and influences the chlorophyll synthesis mechanism that is linked to declining chlorophyll content (Zhang et al., 2019). In this current study, we noticed the AMF recovered Zn deficiency that declined in broccoli under Zn deficient conditions. This study also implies that AMF improved plant biomass and photosynthetic efficiency through the coordination of nutrient balance in broccoli.

4.3. AMF enhanced nutrient accumulation in broccoli

Zn deficiency disturbs mineral homeostasis in plants (Molnár et al., 2022; Strenkert et al., 2023). Zn and Fe often interact with each other and are reported to uptake in plants through a common ion transporter *IRT* gene, though Zn is regulated in plants by ZRT/IRT-like protein (*ZIP*) family genes (Rahman et al., 2020; Tiong et al., 2015). In this study, Zn deficiency led to a decline in Zn content along with Fe concentration in broccoli. This finding suggests that Zn deficiency caused a nutritional imbalance in plants while plants are not able to boost Zn under Zn deficient conditions. Further, we noticed that AMF simultaneously increased Zn and Fe in broccoli, this is possibly achieved through the mycorrhizal association that helps to nutritional flow to the host plants. AM fungi provided a sufficient amount of Zn in broccoli, this evidence supported the upregulation of the *BoZIP1* gene that is highly expressed in response to AMF under Zn deficiency. The *ZIP1* is a key Zn transporter that reported to the coordination of Zn uptake in plants from the soils (Krishna et al., 2020; Claus and Chavarría-Krauser, 2012). As a consequence of Zn and Fe uptake in broccoli, we checked the responses of candidate genes *BoNRAMP1* and *BoMTP1* associated with transporting several metals including Zn and Fe (Castaings et al., 2016; Ricachenevsky et al., 2013). In this current study, we noticed that *BoMTP1* did not exhibit any considerable expression under Zn deficiency, though slightly upregulated in response to AMF under Zn deficiency. This result suggests that the *BoMTP1* gene is not involved in low Zn uptake but rather is involved by AMF associated with Zn deficiency. In contrast, the upregulation of Fe transporter *BoNRAMP1* in combined treatment (Zn deficiency + AMF) and single AMF treatment indicates the sufficient bioavailability of Fe that is typically absorbed into root cells by *BoNRAMP1* in broccoli plants. These above findings suggest that AMF not only alleviates Zn deficiency but also restores Fe status in broccoli.

The elevation of S-containing metabolites (Cys and GSH) in association with AMF under Zn deficiency in broccoli, indicates the hypothesis related to increased S is correlated with the induction of S-containing metabolites in broccoli. We noticed that upregulation of Cys and GSH due to a combination of AMF under Zn deficiency, not only alleviated Zn deficiency but also enhanced antioxidant defense in broccoli. Interestingly, we did not observe any induction of Met, one of the products downstream in the metabolic pathways of Cys (Hesse and Hoefgen, 2003). It is likely that, since Met synthesis is also involved in the processing of other intermediate metabolites, AMF supplementation did not influence these molecules that could ultimately induce Met synthesis in the roots of broccoli. In plants, the roles of Cys and GSH play as antioxidant and/or metabolic precursors of antioxidants and cofactors that enhance plant immunity under cellular stress (Rahman et al., 2020; Álvarez et al., 2012). AMF symbiosis was found to influence S nutrition in *Medicago truncatula* (Sieh et al., 2013). Furthermore, the restoration of Zn due to AMF is one of the key findings of this study. Plant-specific functions such as restoration of PSII, regulation of hormones, and stabilization of CO_2 status in mesophyll tissue are dependable on Zn status (Gupta et al., 2016). Therefore, we noticed that AMF was fully active in alleviating Zn deficiency and boosting S-containing plant defense in broccoli.

4.4. AMF alleviated cellular stress by ROS scavenging

Zn deficiency induced oxidative stress in plant cells (Molnár et al., 2022), and oxidative stress-induced cellular damage caused by ROS in plants (Frei et al., 2010). However, plants have evolved a series of antioxidant enzymes that are effective in alleviating cellular damage caused by ROS in Zn-deficient plant genotypes (Frei et al., 2010). In this present study, AMF supplementation under Zn deficiency significantly improved CAT and APX activity in broccoli roots, suggesting the potential of antioxidants in ROS ($O_2^{\cdot-}$ H_2O_2) homeostasis in roots. CAT decomposes H_2O_2 from cells, and APX leads the conversion of H_2O_2 to H_2O (Li, 2023). Further, the SOD and POD alleviated especially in

combination with AMF with Zn deficient condition in roots. In plants, oxidative stress homeostasis is achieved by antioxidative systems (Raza et al., 2022). These defense systems are triggered by several enzymatic scavengers such as SOD, CAT, POD, and APX. Among these enzymes, SOD played a role as a front liner of stress defense against ROS and was involved in $O_2^{\bullet-}$ dismutation (Haque et al., 2021). However, the increased activity of SOD and POD is due to AMF found in roots, not in shoots of broccoli. It implies that the potentialities of AMF may vary for boosting antioxidants in plant organs. In our investigation, we noticed that increased activity of SOD, CAT, POD, and APX in response to AMF supplementations under Zn deficiency, protects against Zn deficiency mediating oxidative stress-induced cellular stress in broccoli.

5. Conclusion

This study implies a mechanistic basis of AMF-mediated Zn-deficiency alleviation in broccoli (Fig. 8). The AMF supplementation led to the improvement of plant biomass, photosynthetic efficiency, and cellular injuries caused by Zn deficiency in broccoli. AM fungi substantially boosted the accumulation of Zn and Fe in broccoli. However, the recoveries of Zn and Fe were mediated by AMF and up-regulation of *BoZIP1* and *BoNRAMP1* genes in broccoli roots. This study further discloses that AMF showed antioxidative defense against Zn deficiency involving oxidative stress-induced cellular injury in broccoli. These findings explore new windows of AMF-mediating restoration of Zn deficiency in broccoli. The eco-friendly utilization of AMF may encourage biofertilizer industries and farmers to formulate microbiome-based agricultural solutions for organic and sustainable vegetable production. Also, AMF inoculants may become a promising strategy for crop breeders and researchers aiming to improve the performance of oilseed crops.

CRediT authorship contribution statement

Md Arifuzzaman: Writing – original draft, Visualization, Validation, Methodology, Investigation. **Ruby Bagchi:** Visualization, Validation, Methodology, Investigation. **Md Rokibul Hasan:** Methodology, Investigation. **Md Atikur Rahman:** Validation, Investigation, Writing – review & editing. **Md Mostafizur Rahman:** Supervision, Resources. **M. Atikur Rahman:** Writing – review & editing, Validation. **Ahmad Humayan Kabir:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.stress.2024.100415.

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